

María González Rodríguez

glezrodriguezmaria@gmail.com | +34 619 633 497 | LinkedIn | Portfolio

Aerospace Engineer and Industrial Mathematics master's student with experience applying modelling, simulation and optimisation to engineering problems, with a particular interest in R&D across space, defence and robotics.

Experience

- Simulation Intern, ESA Spaceship Poland – Poznań, Poland** Feb 2026 – Jul 2026
Developed a science-parameterised lunar terrain generation module in Unreal Engine 5 for a ROS 2 integrated rover simulation framework, modelling craters, ejecta, degradation and terrain dependent rock distributions.
- Systems Engineer, ISDEFE – Madrid, Spain** Mar 2025 – Jan 2026
Reviewed requirements and design documentation for safety-critical ATM systems and supported V&V through assessment of test plans.
- ATM-CNS Engineer, INECO – Madrid, Spain** Mar 2024 – Mar 2025
Performed conformity studies for aeronautical radio-communication infrastructure using OUNPPM.
- Analyst, Risk Advisory – Core IT, Deloitte – Madrid, Spain** Sep 2023 – Mar 2024
Analysed security protocols and infrastructure, performed data testing and documented findings against relevant security standards.
- Intern, Asteroid Mining Corporation Ltd – London, United Kingdom** Jul 2022 – Oct 2022
Supported early-stage design of a multi-asteroid exploration mission and updated spacecraft mass, power and cost budgets for feasibility.

Education

- MSc Industrial Mathematics, Universidad Politécnica de Madrid** Sep 2024 – Expected Feb 2027
Coursework in dynamical systems, numerical PDEs, multidisciplinary design optimisation (10/10, highest distinction), reduced-order modelling, stability and inverse problems; hands-on modelling with MATLAB and COMSOL Multiphysics.
- BSc Aerospace Engineering, Universidad Politécnica de Madrid** Sep 2018 – Sep 2023
Coursework in orbital mechanics, flight mechanics, control and optimisation, numerical methods, finite elements, aerodynamics and aeroelasticity. Bachelor's thesis awarded the highest distinction.

Projects

- Scientifically Grounded Lunar Terrain Generation for Rover Simulation** 2026
Developed configurable procedural lunar terrain generation for rover perception and navigation simulation, translating lunar-surface statistics into crater, ejecta and rock distributions for Unreal Engine 5.
- Optimizing Solar Panel Placement on the Moon for Maximum Energy Production** 2025
Built a modular MATLAB simulator coupling solar geometry, DEM-based terrain shadowing, surface temperature and panel efficiency to assess lunar energy-production locations.
- ORBITMiNER: Optimal Routing Biological-Inspired Techniques for Mining of Near-Earth Resources** 2023
Developed genetic-algorithm strategies in Python for dynamic routing between moving Near-Earth asteroid targets.
- Random Positioning Machine for Microgravity Simulation** 2022
Developed and analysed rotational-motion algorithms for two- and three-axis partial-gravity simulation, focusing on gravity-vector behaviour and residual acceleration.

Workshops

- ESA Academy Concurrent Engineering Workshop 2025 — ESA, ESEC-Galaxia, Belgium** Sep 2025
Worked in the AOCS team on a Phase 0 solar-sail Jupiter flyby mission, analysing torques, moments of inertia and actuator options using CDP4-COMET.
- Introduction to Space Mining — Universidad Politécnica de Madrid** Apr–May 2021
Studied space-mining mission design and contributed to a conceptual asteroid-mining mission covering target selection, resource assessment and risk analysis.

Research

- JAMS: Jupiter Analyzer of Magnetosphere with Solar Sailing** Apr 2026
5th Symposium on Space Educational Activities | Contributor | Poster presented
- Scientifically Grounded Lunar Terrain Generation for High-Fidelity Rover Perception and Navigation Simulation** Nov 2026
i-SAIRAS/iSpaRo 2026 | Co-author | Paper submitted, under review
- Development of an Improved Random Positioning Machine to Simulate Organic Growth in Microgravity** Sep 2022
73rd International Astronautical Congress | Co-author | Conference contribution
- Literature Studies and Experimental Characterization of Multiple Solid Propellant Regression Rates Using Crawford Bomb Method** Oct 2020
71st International Astronautical Congress | Co-author | Published conference paper

Skills

- Simulation:** COMSOL Multiphysics, MatCont, Unreal Engine
Engineering Tools: CDP4-COMET, CATIA V5, LaTeX
- Programming:** MATLAB, Simulink, Python, Fortran
Languages: Spanish (Native), English (C1 Advanced)